ASSESSMENT TOOL - 2

SIMATS ENGINEERING

COURSE : ARTIFICIAL INTELLIGENCE 'C'

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ASSESSMENT 2

Hospital Doctor shift Assignment using Backtracking Search

1. Introduction

A hospital needs to assign three doctors (D_1, D_2, D_3) to three shifts: Morning, Afternoon, and Night. The assignment must satisfy all the given constraints. Backtracking Search is used to find a valid assignment

2. Given constraints

1. D_1 cannot work the Night shift.
2. D_2 must work before D_3.
3. Only one doctor can work in each shift.
4. D_3 cannot work the morning shift.
5. Every doctor must be assigned exctly one Shift.

3. Backtracking Search

Step 1: Assign D_1

Possible shift for D_1 are Morning or Afternoon because D_1 cannot work Night.

Choose:

D_1 = Morning

constraint check:

D_1 is not assigned to Night

Step 2 : Assign D_2

D_2 must work before D_3. Therefore, D_2 must have an earlier shift than D_3.

Choose :

D_2 = Afternoon

Constraint check:

* D_2 is assigned before D_3.

Step 3: Assign D_3

The remaining shift is Night

Choose:

D_3 = Night

Constraint check:

D_3 is not assigned to morning

D_2 (Afternoon) is before D_3 (Night).

only one doctor is assigned to each shift.

4. Final Valid Schedule

Shift	Doctor
Morning	D_1
Afternoon	D_2
Night	D_3

Robot Navigation in a Grid using BFS Search

1. Introduction

A robot must move from the start (S) to the final (F) in a n x n grid. It can move up, down, left, right, start moves cell by cell and the robot cannot pass through obstacles.

2. Example Grid

* S: Start
* G: Goal
* #: Obstacle
* .: Free cell.

3 Manhattan Distance Heuristic

Manhattan Distance is calculated as:

$h(n) = |x_1 - x_2| + |y_1 + y_2|$

Where:

* (x_1, y_1) is the current position
* (x_2, y_2) is the goal position

4. Optimal path

The shortest path found by BFS is:

S → Right → Right → Down → Down → Down → Right → Right → Down.

5. Path cost

Each movement costs 1

Number of moves = 9

Total Path cost : 9x9 = 9

6. Conclusion:

BFS explores all reachable cells level by level and finds the shortest path when each movement has the same cost. Obstacles are avoided, and manhatten Distance provides an estimate of the distance to the goal.

3 Autonomous Rescue Robot in a Disaster-Affected Building.

(a) Analyze the Constraints and their effects.

An autonomous rescue robot must reach a survivor while avoiding obstacles and minimizing the total travel cost.

1. Movement Constraints.

The robot can move up, Down, Left, Right. It cannot move diagonally.

Effect: The robot must select a valid direction at every step

Using uniform cost search and online search.

Uniform cost search (UCS) is suitable because the robot's movements have different costs, when risky zones are involved.

Step 1:- Initialize the robot.

* Place the robot at the starting position.
* Set the initial path cost
* Add S to the priority queue.

Step 2: Observe Nearby cells.

The robot senses the cells above, below, left, and right.

Step 3: Calculate movements costs.

* Normal cell movement = 1
* Risky cell movement = 1 + 2 = 3
* Blocked cell movement = Not allowed

Step 4: Update the priority queue.

It adds the cells to the priority.

Step 5: Select the lowest - cost path.

The robot selects the cell with the lowest accumulated cost and moves towards it.

2. Obstacles

Some rooms or paths are blocked and cannot be traversed.

Effect: the robot must avoid blocked cells and search for another route.

3. Limited Sensing

The robot can observe only the cells directly next to its current position.

Effect: It cannot know the complete building layout in advance. It must explore and collect information as it moves.

4. Movement cost.

Each movement has a cost of 1.

Effect: The robot should avoid unnecessary

5. Risky Zones

Entering a risky zone adds an extra cost of 2

Therefore, the total cost of entering risky zone is 3.

Effect: The robot should prefer a safer route if it has a lower total cost.

6. Dynamic knowledge

The robot must update its knowledge whenever it it has a lower total cost.

Effect: It must change its plan when new information becomes available

ALGORITHM

1. Start at S with cost 0
2. Observe neighbour cells
3. Ignore blocked cells
4. Calculate the cost of each valid move
5. Select the path with lowest total cost.
6. Update the map using new observations
7. Repeat until G is reached.

(c) Application of AI concepts.

1. Intelligent Agents.
   * The robot senses nearby cells.
   * It decides Obstacle
   * It selects a path
2. Rationality.

It means selecting the best possible action based on the available information

(d) Justification for choosing uniform cost search and online search.

Uniform cost search is suitable because it selects ~~because~~ ~~it~~ ~~sel~~ path with the lowest accumulated cost. UCS can compare different routes and choose a less expensive one.

Advantages:

1. It considers both normal and risky movement costs.
2. It avoids blocked cells.
3. It helps find a minimum-cost path when the relevant map and costs are known.
4. Online exploration allows the robot to respond to new information.
5. It supports rational decision-making in uncertain environment.